DrumFeatureDemonstrator Documentation

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October 7, 2026, for DrumFeatureDemonstrator 0.0.1

1 Introduction

DrumFeatureDemonstrator is a one-sample drum player with a step grid (1/16 notes, 8 to 32 steps). You load one sample (a drum hit, a cymbal, a reversed cymbal), choose the steps on which it is played, and change pitch, direction and loudness per step. The sample is also played from MIDI notes.

The new thing is **play to end**. The plug-in knows exactly how long the sample sounds at the chosen pitch. A step in the mode *reverse + play to end* is therefore started early, so that the sample *ends exactly at the end of that step*. Put it on the last step before a beat, and a reversed cymbal leads into the next beat without any manual nudging, at any pitch and any tempo. The plug-in reads the beat position from the DAW, so it always stays in sync.

2 Installation

The downloaded file is a zip file. It contains the plug-in (.vst3; on macOS also the AU, .component), the Standalone application, a ReadMeFirst.txt, the licence files and this manual. Copy the plug-in to the right place and let your DAW rescan its plug-ins.

```
Windows (VST3):  C:\Program Files\Common Files\VST3
Mac (VST3):      /Users/yourUSERNAME/Library/Audio/Plug-Ins/VST3
Mac (AU):        /Users/yourUSERNAME/Library/Audio/Plug-Ins/Components
Linux (VST3):    /home/yourUSERNAME/.vst3
```

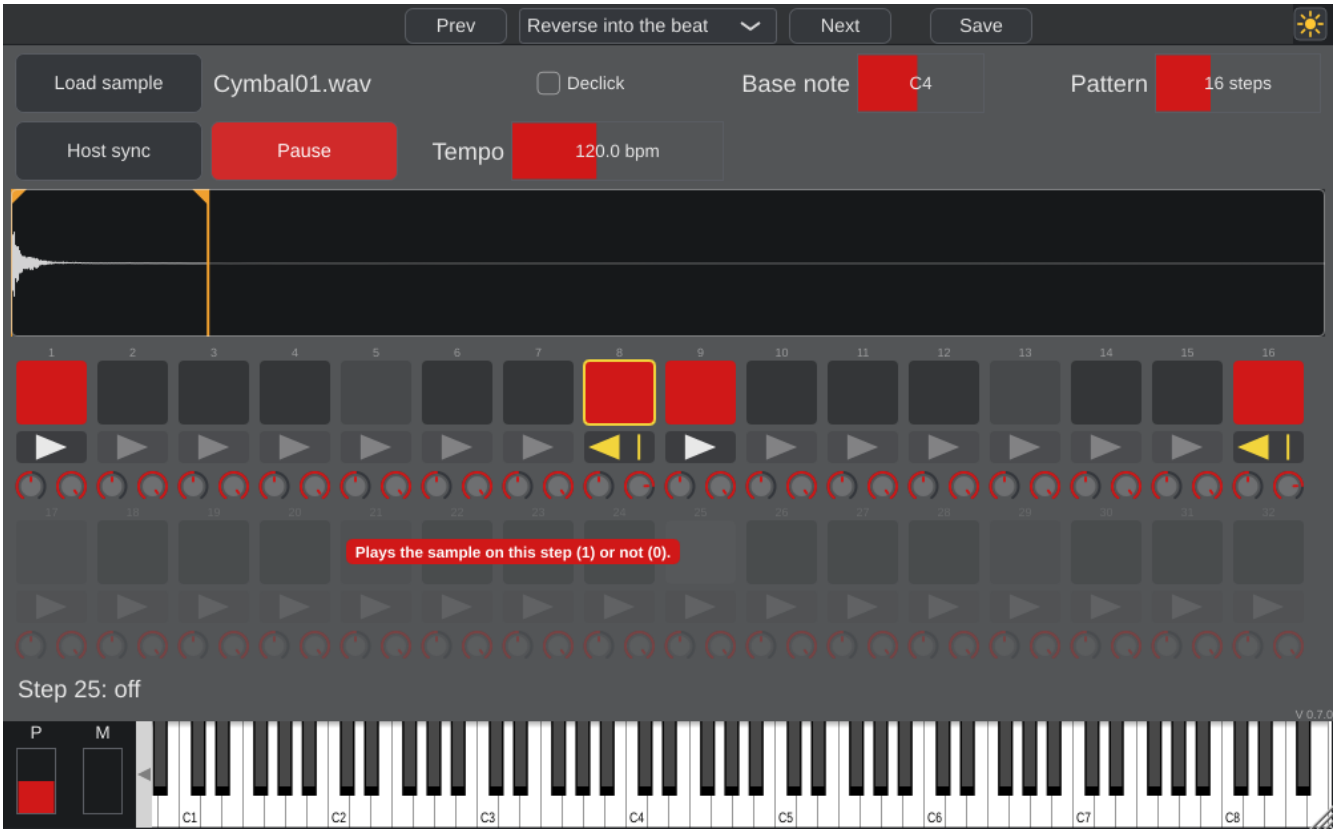
The macOS version is not signed with an Apple Developer ID. If macOS reports that the plug-in is damaged or cannot be opened, remove the quarantine flag in the Terminal, e.g. `xattr -cr ~/Library/Audio/Plug-Ins/VST3/DrumFeatureDemonstrator.vst3`.

2.1 Presets and settings

The presets (plain XML files) and the settings file (`user.settings`: day/night theme) are stored in this folder – delete it to remove DrumFeatureDemonstrator completely:

```
Windows: C:\Users\yourUSERNAME\AppData\Roaming\Jade_Hochschule\DrumFeatureDemonstrator
Mac:     /Users/yourUSERNAME/Library/Audio/Presets/Jade_Hochschule/DrumFeatureDemonstrator
Linux:   /home/yourUSERNAME/.config/Jade_Hochschule/DrumFeatureDemonstrator
```

3 How to use



1. **Load sample**: choose a sample file (wav, aiff, flac, ogg), or drop a file onto the waveform. A copy is kept in the folder `Samples` next to the presets; the sample is stored with your DAW project and referred to by its name in a preset.

2. **Waveform**: drag the two orange handles to set start and end of the part that is played. With **Declick** switched on, a handle snaps to a zero crossing of the sample (no click at the edge).

3. **Base note**: the MIDI note at which the sample sits in the scale, i.e. plays at its original speed (C4 = MIDI note 60). Changing it transposes all steps and the keyboard. MIDI notes (also the keyboard at the bottom) play the sample with the pitch relative to the base note and with velocity.

4. **Pattern**: length of the pattern in steps (8 to 32). Steps beyond the length are dimmed.

5. **Host sync**: on = the plug-in follows tempo and position of the DAW (it plays while the DAW plays); off = internal beat with its own tempo and the **Play/Pause** button (or the **space key**).

6. **Steps**: per step a button (sample on/off), below it the **mode button** (click: forward → reverse → reverse + play to end → forward), and two small knobs: left the note (a MIDI note, shown as e.g. C4; the sample sounds at its original pitch when the note equals the base note), right the velocity (MIDI velocity 1 to 127). The orange frame shows the step that is playing.

7. **Info line** (below the steps): shows the value of the control under the mouse and of the control you change, e.g. "Step 5 note: E4 (MIDI note 64)" – the knobs have no edit fields.

8. **Preset handler** and **day/night**: as usual; Save turns red when you change a control. Presets contain the pattern, the region and the name of the sample. Factory presets keep the sample you have loaded.

3.1 Selecting a step and entering notes

While the transport stands (the DAW is stopped, or the internal beat is paused) the *selected step* is framed in yellow; while the plug-in plays, the frame follows the playing step. Click into a step (or use the cursor keys: left/right ±1 step, up/down ±16 steps, Home/End first/last step) to select it. A **MIDI note** (from the keyboard at the bottom of the window, or from your MIDI keyboard) sets the **note and the velocity** of the selected step and switches the step on; the sample is also heard. The selection does not move by itself; use the cursor keys to go to the next step.

3.2 Play to end: what to know

- The sound ends at the *end of the step* on which the mode is set. To lead into beat 1, set it on step 16 of a pattern of 16 steps.
- If the start time has already passed (the transport starts or jumps into the run-up), the sound is *not* played and a notice appears.
- Play to end is only possible if the sample (at its pitch) is not longer than the pattern. Otherwise the mode button is outlined in red and a notice is shown: shorten the sample with the end handle, raise the pitch or lengthen the pattern.
- Tempo changes during the run-up are not followed.
- Forward + play to end is not offered; reverse the sample file beforehand if you need it.

4 Controls

All controls, their range and default value. Rest the mouse on a control to see its description as a tooltip. Every step n (1 to 32) has four parameters (`Step`*nn*`Active`, `Mode`, `Note`, `Velocity`); they are described once in the last rows of the table.

Control	Range	Default	Description
Base note	0 - 127	60 (C4)	MIDI note at which the sample plays at its original speed.
Start	0 - 1	0	Start of the played region, as a fraction of the sample length (waveform handle).
End	0 - 1	1	End of the played region (waveform handle).
Pattern length	8 - 32 steps	16	Number of 1/16 steps of the pattern.
Sync to host	off / on	on	On: follow the tempo and position of the DAW. Off: internal tempo and play button.
Internal tempo	40 - 240 bpm	120	Tempo of the internal beat.
Internal play	pause / play	pause	Plays or pauses the internal beat (only without host sync).
Declick	off / on	off	Handles snap to a zero crossing of the sample when moved.
Step n on	off / on	off	Plays the sample on this step.
Step n mode	0 / 1 / 2	0	0 forward, 1 reverse, 2 reverse and play to end.
Step n note	12 - 108 (C0 - C8)	60 (C4)	MIDI note of the step; the sample sounds at its original pitch when the note equals the base note.
Step n velocity	1 - 127	100	MIDI velocity of the step (volume = velocity / 127).

5 How it works

For a step in the mode reverse + play to end the plug-in computes the *duration* of the sample in host samples,

$$d = \frac{N}{r}, \qquad r = 2^{p/12}\,\frac{f_{\text{file}}}{f_{\text{host}}},$$

with the number of samples N of the played region, the pitch p in semitones (the note of the step minus the base note) and the two sample rates. The start time is the end of the step minus d. Because d is at most one pattern long, the start lies at most one pattern pass before the step, so the plug-in looks ahead one pass. The resampling is done by linear interpolation; pitch and duration change together.

The drum samples used for the examples and tests come from MusicRadar (SampleRadar, free and royalty-free): https://www.musicradar.com/news/tech/free-music-samples-royalty-free-loops-

6 Source code

DrumFeatureDemonstrator is open source. You find the source code here: https://github.com/JoergBitzer/DrumFeatureDemonstrator. Please report bugs there, too.

7 Legal stuff

For this documentation the CC-BY 4.0 licence is valid. Copyright holder is Joerg Bitzer. The source code of the plug-in is under the MIT licence. The plug-in binaries contain JUCE (used under the AGPLv3), the VST3 SDK (MIT) and, on macOS, the Audio Unit SDK (Apache 2.0); the binaries as a whole are distributed under the AGPLv3.

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